EPICHUB ILLUSTRATED REFERENCE & WORKBOOK

MATH, FRACTIONS, DECIMALS, RATIOS, PERCENTAGES & VEDIC SPEED CALCULATION

Complete Reference Guide, Illustrated Textbook & Cheat-Sheet Workbook — Classical Number Theory Meets High-Speed Indian Mental Arithmetic. **Authored by EPICHUB.**



16 SUTRAS	1/2 TO 1/25	2 TO 13	25 DRILLS
VEDIC MENTAL ALGORITHMS	MASTER RECIPROCAL GRID	OSCULATOR DIVISIBILITY	TIMED SPEED WORKBOOK

UNIT 01: CORE NUMBER SENSE & ARITHMETIC FOUNDATIONS

FOUNDATIONS & PROOFS



ARCHITECTURAL NUMBER SENSE & DIGITAL ROOT VERIFICATION

Every rapid mental calculation rests on structural fluency with prime factorization, algebraic identities, and modular casting-out-nines (*Beejank*). Before executing multi-digit shortcuts, a practitioner inspects digit sums and unit digits to verify outputs in under two seconds.

Concept / Law	Algebraic / Symbolic Form	Tactical Speed Application & Example
Distributive Split	$a \times (b \pm c) = ab \pm ac$	Break awkward multipliers into base-10 anchors: $47 \times 98 = 47 \times (100 - 2) = 4700 - 94 = 4606$.
Difference of Squares	$a^2 - b^2 = (a - b)(a + b)$	Multiply equidistant numbers around a midpoint: $76 \times 84 = (80 - 4)(80 + 4) = 80^2 - 4^2 = 6400$.
GCD × LCM Identity	$\text{GCD}(a, b) \times \text{LCM}(a, b) = a \times b$	For $a = 24, b = 36$: $\text{GCD} = 12$, so $\text{LCM} = (24 \times 36) \div 12 = 2 \times 36 = 72$.
Number of Factors	If $N = p^a \times q^b \times r^c$, then $d(N) = (a+1)(b+1)(c+1)$	For $360 = 2^3 \times 3^2 \times 5^1$, total positive divisors = $(3+1)(2+1)(1+1) = 4 \times 3 \times 2 = 24$.
Digital Root (Beejank)	$DR(N) = N \bmod 9$ (where $9 \equiv 0$)	Verify $43 \times 26 = 1118$: $DR(43)=7$, $DR(26)=8$; $DR(7\times8)=DR(56)=2$. $DR(1118)=DR(11)=2$. Verified!

High-Speed Divisibility Diagnostic Matrix (Divisors 2 through 13)

Divisor	Diagnostic Rule	Rapid Verification Example
2, 4, 8	Last **1**, **2**, or **3** digits are divisible by 2^1 **(2)**, 2^2 **(4)**, or 2^3 **(8)** respectively.	In **57,312**: **312 ÷ 8 = 39** (exact), so divisible by **2, 4,** and **8**.
3 & 9	Sum of digits (or casting out 9s) is a multiple of **3** or **9**.	**84,519 → 8+1=9, 4+5=9, 9**; remainder **0**, so divisible by **3** and **9**.
6 & 12	For **6**: passes both **2** and **3**. For **12**: passes Coprime pair **3** and **4**.	**1,728**: ends in **28** (divisible by **4**) and digit sum **18** (divisible by **3**) → divisible by **12**.
7 (Osculator)	Double the unit digit and subtract from the remaining truncated prefix: **Prefix − 2 × Unit**.	**658 → 65 − (2 × 8) = 49**. Since **49 = 7 × 7**, **658** is divisible by **7**.
11	Absolute difference between sum of odd-place digits and even-place digits is **0** or **11k**.	**91,839 → (9+8+9) − (1+3) = 26 − 4 = 22** (multiple of **11**).
13 (Osculator)	Multiply the unit digit by **4** and add to the remaining truncated prefix: **Prefix + 4 × Unit**.	**728 → 72 + (4 × 8) = 72 + 32 = 104 → 10 + (4 × 4) = 26** (**13 × 2**).

UNIT 02: FRACTIONS MASTERCLASS & VISUAL PARTITION MODELS

BUTTERFLY CROSS-METHOD

A fraction **a / b** (**b ≠ 0**) represents both a part-to-whole ratio and an unexecuted division quotient. Rather than computing slow lowest common denominators (LCD) for every two-fraction operation, use the **Butterfly Cross-Multiplication Method** (derived from the Vedic *Urdhva Tiryagbhyam* sutra) to add, subtract, and compare fractions in a single mental sweep.

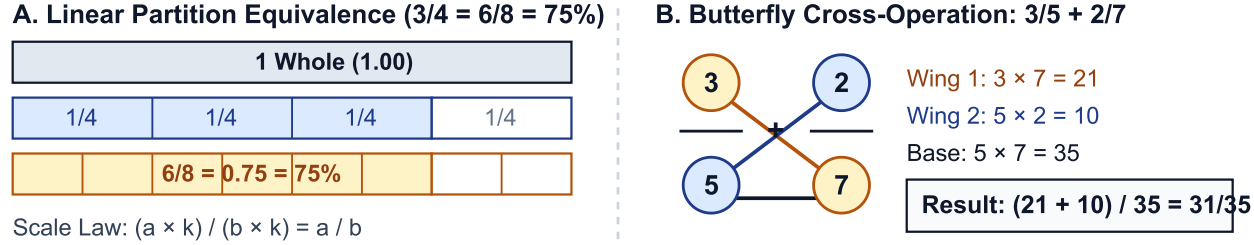


Figure 2.1: Left — Linear bar equivalence showing why scaling numerator and denominator preserves proportional value. Right — The Vedic cross-multiplication butterfly for rapid fraction addition, subtraction, and inequality comparison.

1. CONSTANT-GAP THEOREM

For proper fractions (**a < b**) with constant gap **b − a = k** (e.g., **4/7, 9/12, 29/32**), the largest numerator (**29/32**) is strictly the greatest.

2. SPLIT-WHOLE SUBTRACTION

To compute **100 − 37 5/13**, rewrite **100** as **99 + 13/13**. Subtract wholes (**99−37=62**) and complements (**(13−5)/13**) → **62 8/13**.

3. TELESCOPING SERIES

Use **1/[n(n+1)] = 1/n − 1/(n+1)**. Thus **1/(1×2) + 1/(2×3) + ⋯ + 1/(99×100)** collapses to **1 − 1/100 = 99/100**.

UNIT 03: DECIMALS, REPEATING CYCLES & MASTER CONVERSION GRID

RECIPROCAL CHEAT-SHEET

A simplified fraction **p / q** terminates in base-10 if and only if the prime factorization of denominator **q** contains exclusively powers of **2** and **5** (**$q = 2^m \times 5^n$**). The number of decimal places before termination equals **max(m, n)**. Any other prime factor in **q** produces a repeating decimal block.

RECURRING DECIMAL TO FRACTION CONVERSION LAWS:

1. **PURE RECURRING DECIMAL:** Place the repeating block over as many **9**s as repeating digits: **0.abcabc… = abc / 999**. Example: **0.272727… = 27 / 99 = 3 / 11**.
2. **MIXED RECURRING DECIMAL:** Subtract the non-repeating prefix from the full digits and divide by **9**s (for repeating digits) followed by **0**s (for non-repeating decimal digits): **0.ab(cd)… = (abcd − ab) / 9900**. Example: **0.1666… = (16 − 1) / 90 = 15 / 90 = 1 / 6**.

The EPICHUB Master Reciprocal & Percentage Cheat-Sheet (1/2 to 1/25)

Fraction	Decimal	Exact %	Key Multiples & Cyclic Properties	Fraction	Decimal	Exact %
1 / 2	0.5000	50%	Base binary anchor; **× 0.5 = ÷ 2**	**1 / 12**	0.0833…	8.33% (**8 1/3%**)
1 / 3	0.3333…	33.33% (**33 1/3%**)	**2/3 = 66.67%**	**1 / 13**	0.076923…	7.69% (**7 9/13%**)
1 / 4	0.2500	25%	**3/4 = 0.75 = 75%**	**1 / 14**	0.071428…	7.14% (**7 1/7%**)
1 / 5	0.2000	20%	**2/5 = 40%, 3/5 = 60%, 4/5 = 80%**	**1 / 15**	0.0666…	6.66% (**6 2/3%**)
1 / 6	0.1666…	16.67% (**16 2/3%**)	**5/6 = 0.8333… = 83.33%**	**1 / 16**	0.0625	6.25% (**6 1/4%**)

Fraction	Decimal	Exact %	Key Multiples & Cyclic Properties	Fraction	Decimal	Exact %
1 / 7	0.142857…	14.28% ($14\frac{2}{7}\%$)	Cyclic wheel of **1-4-2-8-5-7** (**2/7 = 28.57%**)	**1 / 17**	0.05882…	5.88% ($5\frac{15}{17}\%$)
1 / 8	0.1250	12.5% ($12\frac{1}{2}\%$)	**3/8 = 37.5%, 5/8 = 62.5%, 7/8 = 87.5%**	**1 / 18**	0.0555…	5.55% ($5\frac{5}{9}\%$)
1 / 9	0.1111…	11.11% ($11\frac{1}{9}\%$)	**k/9 = 0.kkkk…** (e.g., **4/9 = 44.44%**)	**1 / 19**	0.05263…	5.26% ($5\frac{5}{19}\%$)
1 / 10	0.1000	10%	Shift decimal left 1 place	**1 / 20**	0.0500	5.00%
1 / 11	0.0909…	9.09% ($9\frac{1}{11}\%$)	Multiples of 09 (**4/11 = 0.3636… = 36.36%**)	**1 / 24**	0.04166…	4.16% ($4\frac{1}{6}\%$)
3 / 8	0.3750	37.5% ($37\frac{1}{2}\%$)	**37.5% of 64 = (3/8) × 64 = 24**	**1 / 25**	0.0400	4.00%



The Cyclic Seventh Wheel (142857 Magic Ring)

Every proper multiple of 1/7 rotates the exact same 6-digit necklace clockwise:

• 1/7 = 0.142857… • 2/7 = 0.285714… • 3/7 = 0.428571…

• 4/7 = 0.571428… • 5/7 = 0.714285… • 6/7 = 0.857142…

Notice opposite pairs across the wheel always sum to 9: (1+8=9, 4+5=9, 2+7=9) — Vedic Nikhilam sy

Figure 3.1: Visualising the cyclic number **142,857** for instant mental conversion of any seventh fraction (**k/7**).

UNIT 04: RATIOS, PROPORTIONS, ALLIGATION & PERCENTAGE ARCHITECTURE

COMMERCIAL SPEED MATH

A ratio **a : b** expresses relative magnitude and remains invariant under non-zero scalar multiplication (**ka : kb**). When given linked ratios **A : B = m : n** and **B : C = p : q**, combine them via the **Zig-Zag N-Multiplier** into **A : B : C = (m × p) : (n × p) : (n × q)**.

Ratio Transformation	Algebraic Rule (Given)	Speed Application
Componendo & Dividendo	$(a + b) / (a - b) = (c + d) / (c - d)$	Instantly solves $(x + y)/(x - y) = 9/5 \Rightarrow x/y = (9+5)/(9-5)$.
Equal Ratio Sum Law	$a/b = c/d = e/f = (a + c + e) / (b + d + f)$	Any linear combination of numerators over denominators preserves the base ratio k.
Mean Proportional	$a : x :: x : b \Rightarrow x = \sqrt{(ab)}$	Mean proportional between 16 and 36 is $\sqrt{(16 \times 36)} = 4 \times 6 = 24$.
Alligation Rule (Mixtures)	$Q_{cheap} / Q_{dear} = (D - M) / (M - C)$	Mixing \$20/kg and \$35/kg tea to get \$26/kg mean (M): ratio is $(35-26) : (26-20) = 9 : 6 = 3 : 2$.

THE GOLDEN COMMUTATIVE LAW OF PERCENTAGES: $x\%$ **of** $y = y\%$ **of** x

- To calculate 36% **of** 25: flip to 25% **of** $36 = (1/4) \times 36 = 9$.
- To calculate 64% **of** 12.5: flip to 12.5% **of** $64 = (1/8) \times 64 = 8$.
- To calculate 84% **of** 50: flip to 50% **of** $84 = (1/2) \times 84 = 42$.

Percentage Scenario	Master Formula	Worked Tactical Example
Successive Net % Change	$\text{Net \%} = a + b + (a \times b) / 100$	Price rises 20% then 15%: $20 + 15 + (300/100) = +38\%$. If $+20\%$ then -20%: $-(20^2/100) = -4\%$ loss.
Product Constancy ($A \times B = k$)	Increase by n / d requires decrease by $n / (d + n)$	Fuel price rises 25% $(+1/4)$; to keep budget flat, cut consumption by $1/(4+1) = 1/5 = 20\%$.
Markup & Discount Net Profit	$\text{Profit \%} = M - D - (M \times D) / 100$	Goods marked up 40% and sold at 15% discount: $40 - 15 - (600/100) = 25 - 6 = +19\%$ gain.
72-Rule (Compound Doubling)	$\text{Years to Double} \approx 72 \div \text{Rate (\%)}$	At 9% annual compound interest, capital doubles in $\approx 72 \div 9 = 8.0$ years.

UNIT 05: VEDIC SPEED CALCULATION — SUTRAS & VISUAL BLUEPRINTS

16 SUTRAS MASTERY



ANCIENT INDIAN SUTRAS FOR SINGLE-LINE MENTAL ARITHMETIC

Systematized by Bharati Krishna Tirthaji into 16 *Sutras* (aphorisms) and 13 *Upa-Sutras*, Vedic Mathematics replaces rigid right-to-left multi-line scratchwork with holistic pattern recognition—allowing 2-digit, 3-digit, and base-1000 multiplications in a single mental line.

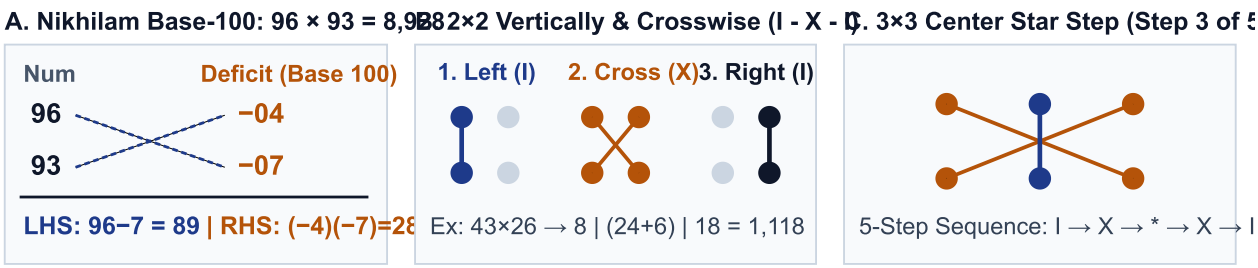


Figure 5.1: Left — Nikhilam Base-100 deviation cross-subtraction. Middle & Right — Urdhva Tiryagbhyam dot-matrix patterns for 2-digit (**I–X–I**) and 3-digit (**I–X–*–X–I**) single-line multiplication.

Master Catalogue of Vedic Speed Sutras & Worked Blueprints

Vedic Sutra / Method	Condition / Trigger	Step-by-Step Worked Blueprint
1. Ekadhikena Purvena ("By One More than the Previous One")	Squaring numbers ending in **5**; or multiplying **ab** × **ac** where **b + c = 10**.	• **Squaring 85^2:** Multiply prefix **8** by **(8 + 1) = 9** → **72**; append $5^2 = 25$ → **7,225**. • **Complementary Units (63 × 67):** Tens match (**6**) and units sum to **10 (3+7)** → **6 × (6+1) | (3 × 7) = 42 | 21 = 4,221**.
2. Nikhilam (Above Base) Surplus Base Multiplication	Both numbers slightly above **100** or **1000**.	• **Compute 108 × 112 (Base 100):** Surpluses are **+08** and **+12**. LHS: **108 + 12 = 120**. RHS: **08 × 12 = 96**. Concatenate → **12,096**.
3. Nikhilam (Mixed Base) One Above, One Below Base	One multiplier > **B**, one multiplier < **B**.	• **Compute 106 × 94 (Base 100):** Deviations **+06** and **−06**. LHS: **106 − 6 = 100**. RHS: **(+6)(−6) = −36**. Borrow **1** (= **100**) from LHS: **99 | (100 − 36) = 9,964**.
4. Ekanyunena Purvena ("By One Less than the Previous")	Multiplying any number by a string of **9**s (**99, 999, 9999**).	• **Compute 647 × 999:** LHS: Subtract **1** from multiplicand → **647 − 1 = 646**. RHS: Apply Nikhilam complement to **647** (**999 − 646**) → **353**. Result → **646,353**.

Vedic Sutra / Method	Condition / Trigger	Step-by-Step Worked Blueprint
5. Multiplication by 11 & 111 Sliding Window Addition	Multiplying any multi-digit integer by **11**.	• **Compute 4,523 × 11:** Keep outer digits and add adjacent pairs right-to-left: **4 | (4+5) | (5+2) | (2+3) | 3 = 49,753**.
6. Yavadunam Tavadunikritya Base Squaring Algorithm	Squaring any number near a base **B** or sub-base **k × B**.	• **Formula: $N^2 = (N \pm d) \mid d^2$**. • **Compute 92^2 (d = −8):** **(92 − 8) | $(-8)^2$ = 84 | 64 = 8,464**. • **Compute 113^2 (d = +13):** **(113 + 13) | 13^2 = 126 | $_{1}69$ = 12,769**.
7. Duplex Squaring (D-Method) *Dwandwa Yoga* Universal Square	Squaring any arbitrary 2-digit or 3-digit integer.	• For **ab**: **D(a) = a^2**, **D(ab) = 2ab**. Thus **$(ab)^2 = a^2 \mid 2ab \mid b^2$**. • **Compute 74^2:** **7^2 | 2(7)(4) | 4^2 = 49 | 56 | 16 → 49 | 57 | 6 → 5,476**.
8. Paravartya & Nikhilam Division Single-Line Synthetic Division	Dividing by **9**, **99**, **89**, or numbers near base **10^k**.	• **Divide 213 ÷ 9:** Split last digit for remainder (**21 | 3**). Running cumulative sum from left: first digit **2**, next **2+1 = 3** (Quotient = **23**); Remainder **2+1+3 = 6**.

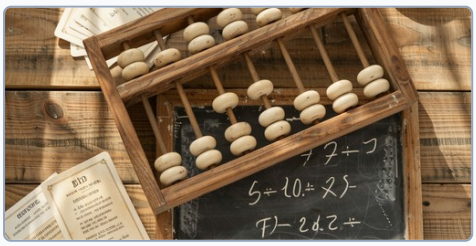
Exact Mental Square-Root & Cube-Root Extraction Charts

When extracting the cube root of a 4-to-6-digit perfect cube (e.g., **$\sqrt[3]{175,616}$**), strike the last three digits (**616**), match the unit digit (**6 → 6**), and find the largest perfect cube ≤ **175** (**$5^3 = 125 \le 175 < 6^3$**, so tens digit is **5**). Thus **$\sqrt[3]{175,616} = 56$** instantly!

Root Unit Digit	0	1	2	3	4	5	6	7	8	9
Square (n^2) Unit Digit	0	1	4	9	6	5	6	9	4	1
Cube (n^3) Unit Digit (1-to-1)	0	1	**8**	**7**	4	5	6	**3**	**2**	9
Exact Cube (n^3) Value	0	1	8	27	64	125	216	343	512	729

UNIT 06: CHEAT-SHEET WORKBOOK — 25 TIMED SPEED DRILLS

TARGET TIME: 8 MINS



MENTAL SPEED-DRILL PROTOCOL (NO CALCULATOR PERMITTED)

Solve all 25 problems using the mental shortcuts and Vedic Sutras taught in Units 1–5. Record your intermediate **LHS | RHS** partitions directly on the answer lines and verify your score against the Annotated Answer Key.

Section A: Fractions, Decimals, Ratios & Percentage Speed Drills (Questions 1–12)

#	Problem Statement & Tactical Hint	Mental Workspace & Final Answer
01	Compute $4/7 + 3/11$ using the Butterfly Cross-Multiplication method.	Ans: _____
02	Subtract $500 - 142\ 7/19$ mentally using complement splitting ($499 + 19/19$).	Ans: _____
03	Determine which fraction is strictly larger: $13 / 17$ or $19 / 25$.	Ans: _____
04	Convert the mixed recurring decimal $0.4363636\ldots$ (0.436 with 36 repeating) to its simplest fraction.	Ans: _____
05	Express $5 / 7$ as a 6-digit repeating decimal using the Cyclic Seventh Wheel.	Ans: _____
06	Compute 37.5% of $480 + 14.28\%$ of 350 using exact reciprocal fractions.	Ans: _____
07	Apply Percentage Symmetry ($x\%$ of $y = y\%$ of x) to evaluate 72% of 12.5.	Ans: _____
08	If $A : B = 2 : 3$ and $B : C = 4 : 5$, find the combined ratio $A : B : C$.	Ans: _____
09	Given $(x + y) / (x - y) = 13 / 5$, find $x : y$ via Componendo & Dividendo.	Ans: _____
10	A stock rises by 30% in Year 1 and falls by 20% in Year 2. Find the net percentage change.	Ans: _____
11	Commodity prices jump by 20% $(+1/5)$. By what exact % must usage drop to keep expenditure unchanged?	Ans: _____
12	In what ratio must 15% acid solution be mixed with 40% acid solution to yield 25% concentration?	Ans: _____

Section B: Vedic Speed Arithmetic & Root Extraction Drills (Questions 13–25)

#	Problem Statement & Target Vedic Sutra	Mental Workspace & Final Answer
13	Subtract $100{,}000 - 64{,}283$ via *Nikhilam* ("All from 9, last from 10").	Ans: _____
14	Multiply 94×91 using *Nikhilam* Base-100 Deficit Method.	Ans: _____
15	Multiply 107×111 using *Nikhilam* Base-100 Surplus Method.	Ans: _____
16	Multiply 105×93 using *Nikhilam* Mixed-Deviation Base-100 Method.	Ans: _____

#	Problem Statement & Target Vedic Sutra	Mental Workspace & Final Answer
17	Compute 994×988 using *Nikhilam* Base-1000 Method.	Ans: _____
18	Compute 115^2 and 78×72 using *Ekadhikena Purvena*.	Ans: _____
19	Multiply 839×999 using *Ekanyunena Purvena*.	Ans: _____
20	Multiply $6{,}274 \times 11$ using the Sliding-Pair Addition rule.	Ans: _____
21	Multiply 54×37 in one line using 2×2 *Urdhva Tiryagbhyam* ($\mathbf{I–X–I}$).	Ans: _____
22	Multiply 213×132 in one line using 3×3 *Urdhva Tiryagbhyam* ($\mathbf{I–X–*–X–I}$).	Ans: _____
23	Square 83^2 using the *Dwandwa Yoga* (Duplex) method $a^2 \mid 2ab \mid b^2$.	Ans: _____
24	Extract the exact cube root of the perfect cube $\sqrt[3]{389{,}017}$ by inspection.	Ans: _____
25	Evaluate the telescoping sum $S = 1/(1\times2) + 1/(2\times3) + 1/(3\times4) + \cdots + 1/(49\times50)$.	Ans: _____

APPENDIX: COMPLETE ANNOTATED ANSWER KEY & STEP-BY-STEP PROOFS

SOLUTIONS & AUDITS

Verify your workbook responses below. Every answer includes the exact mental decomposition pathway so you can audit both accuracy and algorithmic efficiency.

#	Final Answer	Step-by-Step Mental & Vedic Breakdown
01	**65 / 77**	Wings: $(4 \times 11) + (7 \times 3) = 44 + 21 = 65$; Base: $7 \times 11 = 77 \rightarrow 65/77$.
02	**$357\ ^{12}/_{19}$**	Split $500 = 499 + 19/19$. Whole: $499 - 142 = 357$; Fraction: $(19 - 7)/19 = 12/19$.
03	**13 / 17 is larger**	Cross-multiply upward: $13 \times 25 = 325$ vs. $17 \times 19 = 323$. Since $325 > 323$, $13/17 > 19/25$.
04	**24 / 55**	$(436 - 4) / 990 = 432 / 990$. Divide numerator and denominator by $18 \rightarrow 24 / 55$.
05	**0.714285…**	Fifth smallest digit in $\{1, 2, 4, 5, 7, 8\}$ is 7; rotate clockwise around $1\text{-}4\text{-}2\text{-}8\text{-}5\text{-}7 \rightarrow 0.714285\ldots$.
06	**230**	$37.5\% = 3/8$ and $14.28\% = 1/7$. Compute $(3/8) \times 480 + (1/7) \times 350 = 180 + 50 = 230$.
07	**9**	Flip to 12.5% of $72 = (1/8) \times 72 = 9$.

#	Final Answer	Step-by-Step Mental & Vedic Breakdown
08	8 : 12 : 15	$(2 \times 4) : (3 \times 4) : (3 \times 5) = 8 : 12 : 15$.
09	9 : 4	$x / y = (13 + 5) / (13 - 5) = 18 / 8 = 9 / 4$.
10	+4% Net Gain	$\text{Net \%} = +30 - 20 + [(+30)(-20)/100] = 10 - 6 = +4\%$.
11	16.67% ($16\frac{2}{3}\%$)	Price increases by $1/5$, so consumption must fall by $1 / (5 + 1) = 1/6 = 16.67\%$.
12	3 : 2	Alligation cross-differences around mean 25%: $(40 - 25) : (25 - 15) = 15 : 10 = 3 : 2$.
13	35,717	Apply 9-complements to $6, 4, 2, 8$ and 10-complement to $3 \rightarrow 35{,}717$.
14	8,554	Deficits -06 and -09. LHS: $94 - 9 = 85$; RHS: $(-6)(-9) = 54 \rightarrow 8{,}554$.
15	11,877	Surpluses $+07$ and $+11$. LHS: $107 + 11 = 118$; RHS: $7 \times 11 = 77 \rightarrow 11{,}877$.
16	9,765	Deviations $+05$ and -07. LHS: $105 - 7 = 98$; RHS: -35. Borrow 100: $97 \mid (100 - 35) = 9{,}765$.
17	982,072	Base 1000 deficits -006 and -012. LHS: $994 - 12 = 982$; RHS (3 digits): $6 \times 12 = 072 \rightarrow 982{,}072$.
18	13,225 & 5,616	$115^2 = (11{\times}12) \mid 25 = 132 \mid 25 = 13{,}225$. $78 \times 72 = (7{\times}8) \mid (8{\times}2) = 56 \mid 16 = 5{,}616$.
19	838,161	LHS: $839 - 1 = 838$; RHS: $999 - 838 = 161 \rightarrow 838{,}161$.
20	69,014	Right-to-left neighbor sums with carry: $6 \mid (6{+}2) \mid (2{+}7) \mid (7{+}4{=}11) \mid 4 \rightarrow 6 \mid 8 \mid {}_{1}0 \mid 1 \mid 4 = 69{,}014$.
21	1,998	$(5{\times}3) \mid (5{\times}7 + 4{\times}3) \mid (4{\times}7) = 15 \mid 47 \mid 28 \rightarrow 15 \mid {}_{4}9 \mid {}_{2}8 \rightarrow 1{,}998$.
22	28,116	5 steps: $2{\times}1{=}2 \mid 2{\times}3{+}1{\times}1{=}7 \mid 2{\times}2{+}1{\times}3{+}3{\times}1{=}10 \mid 1{\times}2{+}3{\times}3{=}11 \mid 3{\times}2{=}6 \rightarrow 2 \mid 7 \mid {}_{1}$.
23	6,889	$8^2 \mid 2(8)(3) \mid 3^2 = 64 \mid 48 \mid 9 \rightarrow (64 + 4) \mid 8 \mid 9 = 6{,}889$.
24	73	Unit digit 7 flips to 3. Prefix 389 lies between 7^3 (343) and 8^3 (512), so tens digit is $7 \rightarrow 73$.
25	49 / 50 (or 0.98)	Telescopes to $(1/1 - 1/2) + (1/2 - 1/3) + \cdots + (1/49 - 1/50) = 1 - 1/50 = 49/50$.